

Hall Effect in Semiconductor

Richard Khoo

University of California, Berkeley

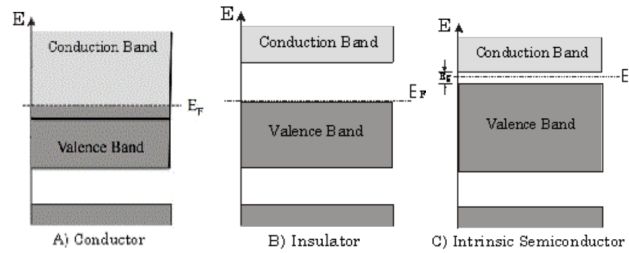
May 6, 2026

Abstract

The Hall Effect and Resistivity of an Al-doped Germanium crystal was measured using van der Pauw geometry while the temperature was changed from 95-350K in various B-fields and currents. From the Hall Coefficient we were able to identify that the material is P-type.

1 Introduction

Measuring the resistivity ρ and the Hall effect of a doped semiconductor shows how charge flows (ρ or σ) and which carriers carry it (indicated in the sign and mag of R_H). In this experiment we used the four-probe Van der Pauw technique on a thin, Al-doped Germanium sample to determine the room-temperature resistivity ρ and the Hall coefficient R_H , then from plotting ρ vs $1/T$ we can identify the extrinsic and intrinsic regions. From ρ and R_H we can find the hall mobility $\mu_H = R_H/\rho$. From the Hall mobility we can calculate the carrier concentration, which semiconductors are measured by. Here we have the energy-band diagrams for a conductor, insulator, and intrinsic semiconductor. The



dark shaded gray parts correspond to the filled states, and the lightly shaded gray parts correspond to unoccupied states. A semiconductor is where there is only a very small energy gap which is to the order of $k_B T$ for such a material there exists a small concentration of empty states (holes) in the valence band and filled states (electrons) in the conduction band due to thermal excitation. As such, the semiconductor has a small but very strongly temperature dependent conductivity.

Furthermore, there exists a doping which is the introduction of a small density of electronic levels at energies within the intrinsically void energy gap, this is done by mixing in some impurity atoms within the semiconductor, these impurity atoms make it so that it either adds extra filled electron levels at energies near the conduction band edge which creates a surplus of electrons in the conduction band or it's used to add extra unfilled levels at energies near the valence band edge. These cases are what define p-type and n-type semiconductors.

2 Theory

2.1 Resistivity (Van dew Pauw)

We first use the Van der Pauw equation (equation 8 from the lab manual):

$$\exp\left(-\frac{\pi d R_{AB,DC}}{\rho}\right) + \exp\left(-\frac{\pi d R_{AD,BC}}{\rho}\right) = 1. \quad (1)$$

by which ρ is implicitly determined and then solving for ρ yields us:

$$\rho = R_s d = \frac{\pi d}{\ln 2} \frac{R_{AB,DC} + R_{AD,BC}}{2} f\left(\frac{R_{AB,DC}}{R_{AD,BC}}\right) \quad (2)$$

2.2 Hall Effect

We used the formula

$$R_H = \frac{V_H d}{IB_z} \quad (3)$$

derived from equation 7 in the lab manual. Here to obtain the Hall Coefficient, we obtained the Hall Voltage by taking the last two sets of measured quantities and then take the average trans-resistance:

$$\frac{R_{AC,BD} + R_{BD,AC}}{2}. \quad (4)$$

According to Ohms law: $V_H = IR_H$ we can obtain

$$V_H = I \frac{R_{AC,BD} + R_{BD,AC}}{2} \quad (5)$$

2.3 Carrier Concentration

Since $I \propto q v_x$ now we know that V_H will depend on whether the charge carriers are mostly electrons or holes. So then by measuring the Hall effect, we can determine whether we have a n or p-type semiconductor. From equation 5 from the lab manual:

$$R_H = \frac{E_H}{J_x B_x} = \frac{1}{en} \quad (6)$$

we can express the carrier concentration in terms of the hall coefficient for electrons (n) and holes (p):

$$n = \frac{1}{q|R_H|}, p = \frac{1}{q|R_H|} \quad (7)$$

2.4 Hall Mobility

The longitudinal transport gives $\sigma = \frac{1}{\rho}$ and combining this with R_H defines the Hall mobility:

$$\mu_H = \frac{|v_d|}{E} \approx R_H \sigma = \frac{R_H}{\rho} \quad (8)$$

3 Apparatus and Procedure

3.1 Overview of the Equipment/Procedure

The procedure starts with cooling the sample to the lowest temperature that we will need, in this case, just about 95K. This cooling is done by adding liquid nitrogen to the crystat as seen in figure 2. Once we reach the desired temperature, a computer interfaced thermometric program is used to control the temperature by using a heater to slowly heat up the sample until the inputted end temperature is reached. This temperature ramp can be terminated at anytime by stopping the experiment. The

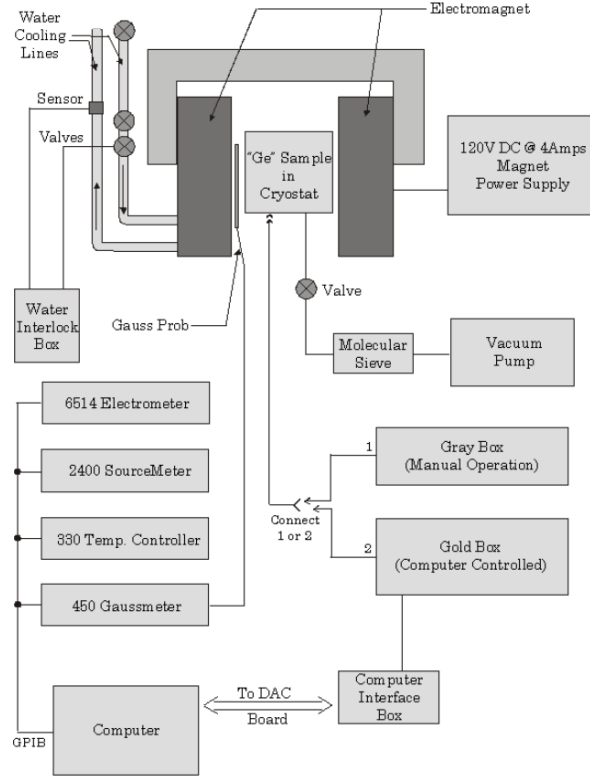


Figure 1: Simple Block Diagram for the Hall Effect in a Semiconductor. (Figure 17 in the lab manual)

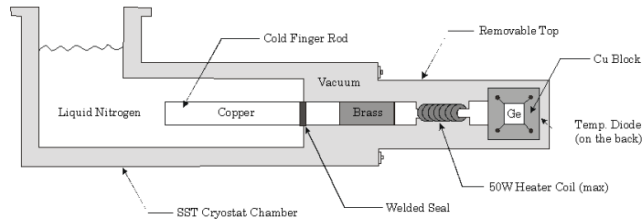


Figure 2: Operating Position of the Cryostat. (Figure 16 in the lab manual)

computer will record data every degree change as the temperature rises at $\approx 5\text{K}$ per minute. There is an upper temperature of 400K , but this will not be used for this experiment. The sample that we used for this experiment is a Al doped germanium crystal that is about $10\text{mm} \times 10\text{mm} \pm 0.01\text{mm}$ with a thickness of about $1.25\text{mm} \pm 0.01\text{mm}$. The germanium crystal is mounted to the front side of the copper block with connection leads A, B, C, and D and the back side reveals a small silicon diode temperature sensor. As seen in Figure 2, there is a brass plug between the LN2 and the sample along with a small air space. This is done such that the sample isn't cooled down too fast nor is it heated up too fast; we want to limit the speed at which the sample heats up or cools down. A vacuum is needed to keep the Germanium sample from oxidizing when heated above room temperature and to cool it easier without the outside environment temperature. In this experiment, we use 5000 Gauss electromagnet to observe the Hall effect. The magnet is powered by 2 60V DC power supplies connected in series, and to prevent the magnet from overheating, water-cooling lines are connected to the magnet as seen in figure 1. The magnet is controlled by the LakeShore 250 Gaussmeter, this provides precise measurements of the magnetic fields. To pass a current through any of the two leads on the sample, we will use the 2400 SourceMeter. To measure the voltage across any two leads, we will use the 6514 Electrometer. The 330 Temperature controller also measures and controls the temperature of the

sample using a silicon diode temperature sensor. The gold box is a Labview software system that automatically triggers the 2400 Source meter to deliver all of the current to the sample leads. In order to connect the current to a pair of leads, it uses a bank of 16 relays to convey this signal.

3.2 Full Procedure

1. Switch on all of the equipment and turn on the Gold Box power.
2. Turn on the water, make sure the Return valve is opened first, then the Main water valve. After this is completed, press the button on the grey interlock box for about 5 seconds.
3. Make sure the vacuum is below 100 milli-torr, after this is completed, turn on the 2 power supplies under the bench. This supplies power to the magnets.
4. Push the Run button on the Gold box and then fill up the cryostat halfway with liquid nitrogen such that the sample cools down to 95K.
5. On the computer, launch the SHE Control Program Front Panel and run the program with a Temperature range of 95-350K, aany B-Field with a maximum of 3500, and a current of about 100-500 μA
6. When the target temperature has been reached, the sample will start to heat up. The temperature ramp is set to increase by 2 Kelvin a minute.
7. Run the procedure using currents ranging from 100-500 μA

4 Calculations and Data Reduction – Analysis

4.1 Plot of ρ of the sample vs inverse temperature

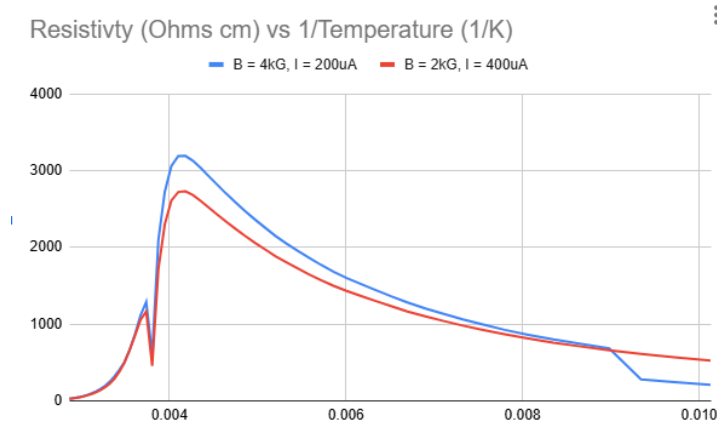


Figure 3: Plot of ρ vs $\frac{1}{T}$

From figure 3 we can see that the material goes from an extrinsic state when the temperature is low to an intrinsic state when the temperature is high. We can see that the turning point is at about 245K when the region turns from an extrinsic to a intrinsic state. We can explain the sharp dip at about 260K due to an anomaly in the system causing a false reading and hence an outlier point.

4.2 Plot of $\sigma = \frac{1}{\rho}$ vs inverse temperature

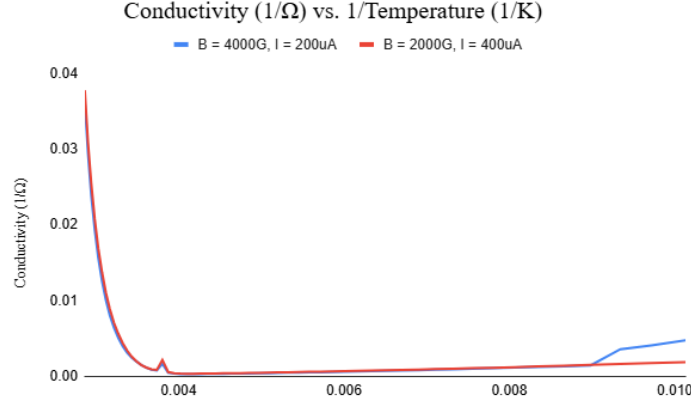


Figure 4: Plot of σ vs $\frac{1}{T}$

From figure 4, we can tell that when the sample is in an extrinsic state, the conductivity is very low but when it changes to an intrinsic state, the conductivity balloons. In the extrinsic region, holes from ionized acceptors dominate with constant density $\rho = N_A$, while the mobility decreases only as a power of temperature. From the Melissinos text, we find the equation 3.19:

$$\mu_H(T) = C_H T^{-3/2} \quad (9)$$

From here we can see the relationship being:

$$\mu_e(T), \mu_h(T) \propto T^{3/2} e^{-E_g/(2k_B T)}$$

and hence:

$$\sigma_{in}(T) = e n_i(T) [\mu_e(T) + \mu_h(T)] \propto e^{-E_g/(2k_B T)}$$

Because the exponential factor dominates as T increases, σ balloons on the left side as temperature gets higher.

4.3 Plot of Hall Coefficient R_H vs inverse temperature

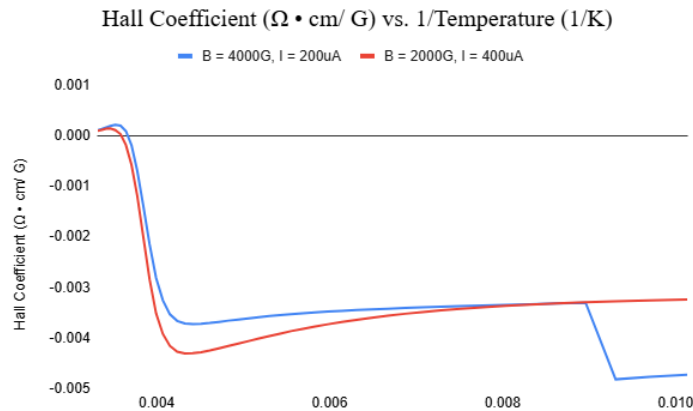


Figure 5: Plot of R_H vs $\frac{1}{T}$

The Hall coefficient for two-carrier is as given in Melissinos eq 3.10 and can also be derived from the equation for the Hall Coefficient $R_H = \frac{E_y}{J_x H}$:

$$R_H = \frac{\mu_h^2 p - \mu_e^2 n}{e(\mu_h p + \mu_e n)^2} \quad (10)$$

As the sample warm, the intrinsic carriers are thermally generated and then eq 10 is necessary. Because $\mu_e > \mu_h$ in Germanium, the electron term grows in weight faster than the hole term, consequently R_H goes toward zero and can change sign. When $R_H = 0$ we can define $b \equiv \mu_e/\mu_h$ and setting $R_H = 0$ in the two-carrier formula gives us the condition:

$$nb^2 = p \text{ at } T = T_0$$

In the intrinsic state of Germanium, electrons and holes are created in pairs:

$$n_i = p_i \propto T^{3/2} e^{-E_g/(2k_B T)}$$

so both n and p rapidly grow and R_H approaches the eq 10 with comparable n and p . This is why $|R_H|$ turns to 0 and then changes signs as T increases.

4.4 Plot Hall Coefficient x Conductivity (Hall Mobility) vs T (where does hall coeff become 0)

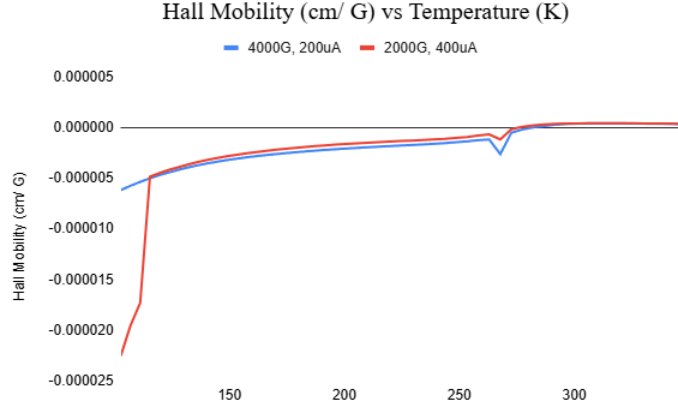


Figure 6: Plot of $R_H \cdot \sigma$ vs T

We plot the Hall mobility which is $\mu_H(T) = R_H(T)\sigma(T)$ versus Time. The curve crosses zero at about $T_0 \approx$ room temperature. The same as in 4.3, R_H at this point equals zero and occurs because of equation 10 and for the same explanation in 4.3. Hence the Hall Coefficient becomes 0 at the inversion temperature T_0 which is at $\approx$ room temperature.

4.5 Carrier concentration and Carrier mobility

The carrier concentration for a p-type material is as follows:

$$R_H = -\frac{1}{ep} = p \approx -\frac{1}{e|R_H|} \quad (11)$$

From here for magnetic field $B = 4000G$ and current $I = 200\mu A$ we get:

$$p = -\frac{1}{eR_H} = \frac{1}{(1.602 \cdot 10^{-19})(7.87 \cdot 10^{-4})} = 7.194 \cdot 10^{21} \text{m}^{-3}$$

where e is the charge of an electron, and the value of R_H is at 300K. The hole mobility is also as follows:

$$\mu_h = \frac{\sigma}{ep} = |R_H|\sigma = \mu_H \text{ (from } \mu_H = R_H\sigma) \quad (12)$$

And if we fit $\mu_h(T)$ to the emperical Ge in the extrinsic region then we get again equation 9:

$$\mu_H(T) = CT^{-3/2}$$

For the mobility ratio we can get extrapolate our conductivity values from the extrinsic region to the point $T = T_0$ and then we obtain the same equation from Melissinos 3.15:

$$b \equiv \frac{\mu_e}{\mu_h} = \frac{R_e(T = T_0)}{R_e(T = T_0) - R_0} = \frac{R_e(T_0)}{R_e(T_0) - R_0} \quad (13)$$

Because $R = \frac{L}{A\sigma}$ we can get:

$$R_0 = \frac{K}{\sigma_0}, R_e(T_0) = \frac{K}{\sigma_e(T_0)}$$

and then substitute into eq 13 and then get:

$$b = \frac{K/\sigma_e}{K/\sigma_e - K/\sigma_0} = \frac{\sigma_0}{\sigma_0 - \sigma_e(T_0)} \quad (14)$$

From here for magnetic field $B = 4000G$ and current $I = 200\mu A$ we get:

$$b = \frac{\sigma_0}{\sigma_0 - \sigma_e(T_0)} = \frac{1.152 \cdot 10^{-3}}{1.152 \cdot 10^{-3} - 2.6 \cdot 10^{-4}} \approx 1.29307 \pm 0.03$$

Where the σ is measured at $T_0 \approx 279$ and $\sigma_e(T_0)$ is extrapolated from an intrinsic fit of $\sigma(T)$ to AT^{-m} , $\sigma_e(T_0) \approx 2.6 \cdot 10^{-4}$. This value is very close to Melissinos's value of $b = 1.31 \pm 0.2$.

4.6 P-type or N-type

In the extrinsic region, the Hall response has a constant sign and magnitude that is set by the majority carriers set by for single-carriers equation 11, A positive hall voltage corresponds to holes which in-turn is p-type. Our Hall-coefficient switch when $R_H - > 0$ has a sign change and because of the two-carrier formula as listed as equation 10, such a change or inversion would only happen in a p-type material.

4.7 Electron/Hole concentration for the sample vs temperature

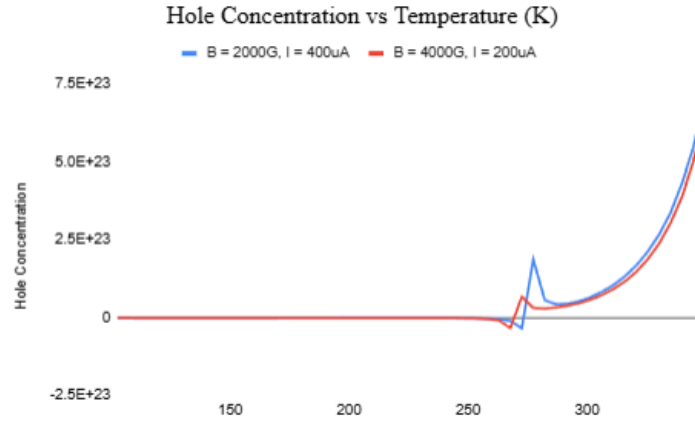


Figure 7: Plot of Hole Concentration vs T

When temperature is low the sample is in an extrinsic state, so holes from the ionized acceptors dominate with rough constant density of $p \approx N_A$. When the sample goes from an extrinsic to an intrinsic state because of the proportion below:

$$n_i = p_i \propto T^{3/2} e^{-E_g/(2k_B T)} \quad (15)$$

we can see that once n_i becomes comparable to N_A , $p(T)$ becomes very large and looks like exponential growth.

4.8 Hall Coefficient R_H and Hall mobility $\mu_H = R_H \cdot \sigma$ vs Temperature

Figure 6 shows our plot for the Hall mobility vs temperature and we found out that at room temperature, and at magnetic field $B = 4000G$ and current $I = 200\mu A$ we get our Hall Coefficient:

$$R_H = 7.87 \cdot 10^{-4}$$

4.9 Hole concentration in the extrinsic region

Calculating the Hole concentration in the extrinsic region uses the same formula as in 4.5 which is equation 11. From $T \approx 140 - 200K$ the Hall coefficient is nearly flat so for where the magnetic field $B = 4000G$ and current $I = 200\mu A$ we will use $|R_H| \approx 3.8 \cdot 10^{-3}$ which is at about $140K$:

$$p \approx \frac{1}{(1.602 \cdot 10^{-19})(3.7 \cdot 10^{-3})} \approx (1.7 \pm 0.1) \cdot 10^{21} \text{ m}^{-3}$$

4.10 Hole mobilities in the extrinsic region

Using equation 12, we can change the temperature to recalculate the hole mobility for in the extrinsic region, we will use the same temperature as that for 4.9 ($\approx 140K$):

First we convert from our table $\sigma = 0.00096165 (\Omega \cdot \text{cm})^{-1} \rightarrow 0.096165 \text{ S/m}$,

$$\mu_h = \frac{0.096165}{(1.602 \cdot 10^{-19})(3.7 \cdot 10^{-3})} = 3.6 \cdot 10^3 \text{ cm}^2/\text{Vs}$$

4.11 Resistance measured for the sample with the magnetoresistance

For the magnetic field $B = 4000G$ and current $I = 200\mu A$ run, we convert the conductivity to resistivity for each block using the following formula:

$$\rho(B) = \frac{1}{\sigma(B)} (\Omega \cdot \text{cm}), B \in \{0, +5kG - 5kG\} \quad (16)$$

From Melissinos figure 3.28 they define the Magnetoresistance as:

$$\frac{\Delta\rho}{\rho_0} \equiv \frac{\rho(B) - \rho(0)}{\rho(0)} = \frac{R_m - R}{R} = MR(B) \quad (17)$$

At $300K$ here we can calculate the ρ values:

$$\rho(0) = 54836\Omega\text{cm}, \rho(+2500) = 89376.23\Omega\text{cm}, \rho(-2500) = 84132.48\Omega\text{cm}$$

From here we can calculate the MR:

$$\begin{aligned} MR(+2500) &= \frac{89376.23 - 54836}{54836} = -0.162 \\ MR(-2500) &= \frac{84132.48 - 54836}{54836} = -0.493 \\ MR_{\text{sym}} &= \frac{\frac{1}{2}(89376.23 - 84132.48) - 54836}{54836} = -0.328 \end{aligned}$$

This makes sense because at room temperature it shows that our MR is negative and at a downwards trend, this reflects the two-carrier nature of Germanium and the temperature-dependent mobilities, so the simple single-carrier positive-MR scaling is not universal. The expected low-field scaling remains at $\propto H^2$ as written in Melissinos.

4.12 Resistivity and Hall Coefficient of the sample at room temperature. At what magnetic field intensity, would the E field due to Hall effect be of the same magnitude (though perpendicular in direction) as the E field due to resistance?

The Hall field in longitudinal field is as follows:

$$E_y = R_H JB, E_x = \rho J = \frac{J}{\sigma}$$

When we set $E_y = E_x$ the required field is as follows:

$$B_{eq} = \frac{\rho}{R_H} = \frac{1}{\sigma R_H} \quad (18)$$

Using the 2kG and 4kG runs we can get a zero field of about:

$$\rho(0) \approx 2.585 \cdot 10^2 \Omega \cdot cm$$

The Hall coefficient is also as follows at room temperature:

$$\begin{aligned} R_H(4 \text{ kG}) &\approx 9.57 \cdot 10^{-5} \Omega \cdot cm/G \\ R_H(2 \text{ kG}) &\approx 1.09 \cdot 10^{-4} \Omega \cdot cm/G \end{aligned}$$

Using the 4kG value of R_H we get:

$$B_{eq} \approx \frac{258.5}{9.57 \cdot 10^{-5}} = 270T$$

Using the 2kG value of R_H we get:

$$B_{eq} \approx \frac{258.5}{1.09 \cdot 10^{-5}} = 238T$$

T is soo large because at room temperature the sample has high ρ while R_H has collapsed towards zero caused by intrinsic mixing of electrons nad holes, so a very big B is required for the hall field to match the longitudinal field.

5 Conclusion

In this experiement we successfully demonstrated the Hall effect in an Al-doped germanium sample. Using the Van dew Pauw method, we were able to determine the resistivity as a function of temperature and observed the transition from extrinsic to intrinsic conduction. At room temperature, the Hall coefficient was positive which confirmed p-type behavior with holes as the dominant carriers. From the measured Hall coefficient and resistivity, we were also able to calculate the carrier denisty on the order of magnitude $10^{21}m^{-3}$ and extracted the Hall mobility, which followed the expected temperature dependence for germanium.

By extending the measurements to higher temperatures we also identified the inversion temperature near 270K where the Hall coefficient approached zero and the sign changed due to the increasing contribution of thermally generated electron-hole pairs. Additionally, analysis of the mobility ratio μ_e/μ_h yielded a value that was consistent with Melissinos validating the experimental and data analysis. Overall we were able to confirm that the Hall effect was a good technique to probe semiconductor transport properties and we were able to determine the sign, concentration, and mobility of carriers, including verifying the distinction between extrinsic and intrinsic conduction regimes.

6 References

1. Melissinos, Adrian. “*The Hall Effect and Properties of Semiconductors.*” *Experiments in Modern Physics*, pp. 80–98. Academic Press (1966). [Book Link](#)
2. A short description of the Hall Effect. [E. Haller Article](#)
3. “Proof of the Van Der Pauw Theorem” [Van Der Pauw Theorem](#)
4. *Hall Effect in a Semiconductor (SHE) Manual*. Physics 111B: Advanced Experimentation Laboratory, University of California, Berkeley. Revised August 27, 2025. [Lab Manual Link](#)

7 Raw Data

As listed in the report guidelines, since our data is unreasonably large to fit on a page, they are instead stored on the graphs that are displayed in this report. However, I will include the graph with 3 measurements but not include it in the report analysis because since we forgot to turn on the magnet, our $B = 2000kG$, $I = 200\mu A$ data for under about 181K is messed up and not usable for analysis.

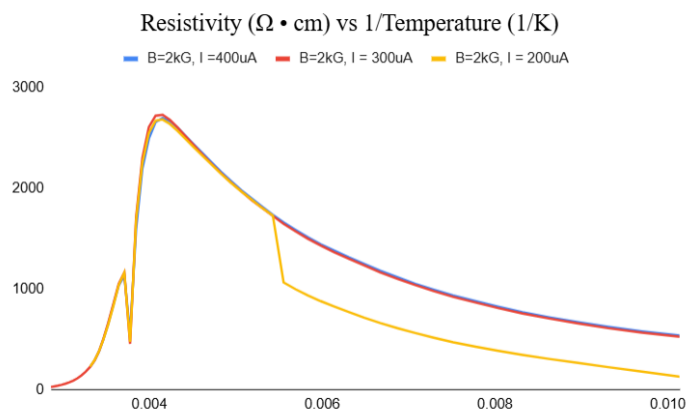


Figure 8: Plot including the $B = 2000kG$, $I = 200\mu A$ dataset